

EXPERIMENT (10)

Ex. No.: 10

Date:

AIM

To create VIEW and INDEX on the Hostel Management System tables.

DESCRIPTION

A VIEW is a virtual table created from a SELECT query. An INDEX improves the speed of data retrieval from a table.

SYNTAX

VIEW

```
CREATE VIEW view_name AS  
SELECT column1, column2  
FROM table_name;
```

INDEX

```
CREATE INDEX index_name  
ON table_name(column_name);
```

```
DROP INDEX index_name ON table_name;
```

Questions on VIEW

- 1. Create a view named v_room containing RoomID, RoomNumber and RentPerMonth from Hostel_Rooms.**

```
CREATE VIEW v_room AS  
SELECT RoomID, RoomNumber, RentPerMonth  
FROM Hostel_Rooms;
```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

191 INNER JOIN Hostel_Warden w
192 ON b.WardenID = w.WardenID;
193 • CREATE VIEW v_room AS
194 SELECT RoomID,
195         RoomNumber,
196         RentPerMonth
197 FROM Hostel_Rooms;
198
199 • SELECT * FROM v_room;

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

RoomID	RoomNumber	RentPerMonth
101	A101	3500.00
102	A102	4500.00
103	B201	5500.00
104	A104	3000.00
105	A105	4500.00
106	B202	4000.00
107	B203	5500.00

Hostel_Warden 835 Hostel_Rooms 836 Result 837 Result 838 Result 839 v_room 840 Read Only

2. Insert a new room through the view and delete the inserted record.

```

INSERT INTO v_room(RoomID,RoomNumber,RentPerMonth)
VALUES(150,'C101',6000);

```

```

DELETE FROM v_room
WHERE RoomID=150;

```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

198 RoomNumber,
199 RentPerMonth
200 FROM Hostel_Rooms;
201 • INSERT INTO v_room
202 (RoomID, RoomNumber, RentPerMonth)
203 VALUES
204 (150,'C101',6000);
205 • SELECT * FROM v_room;
206
207

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

RoomID	RoomNumber	RentPerMonth
101	A101	3500.00
102	A102	4500.00
103	B201	5500.00
104	A104	3000.00
105	A105	4500.00
106	B202	4000.00
107	B203	5500.00
150	C101	6000.00

el_Warden 1294 Hostel_Rooms 1295 Result 1296 Result 1297 Result 1298 v_room 1299 Read Only

The screenshot shows a SQL IDE with multiple tabs. The active tab, 'SQL File 7', contains the following SQL script:

```
200 FROM Hostel_Rooms;  
201 • INSERT INTO v_room  
202 (RoomID, RoomNumber, RentPerMonth)  
203 VALUES  
204 (150, 'C101', 6000);  
205 • SELECT * FROM v_room;  
206 • DELETE FROM v_room  
207 WHERE RoomID=150;  
208 • SELECT * FROM v_room
```

Below the script, the 'Result Grid' displays the output of the last query. It shows a table with three columns: RoomID, RoomNumber, and RentPerMonth. The data is as follows:

RoomID	RoomNumber	RentPerMonth
101	A101	3500.00
102	A102	4500.00
103	B201	5500.00
104	A104	3000.00
105	A105	4500.00
106	B202	4000.00
107	B203	5500.00

The bottom of the IDE shows a tab bar with the following tabs: Hostel_Rooms 1373, Result 1374, Result 1375, Result 1376, v_room 1377, and v_room 1378 (Read Only).

Questions on INDEX

1. Create an index on RentPerMonth and display rooms whose rent is above 4500, then drop the index.

```
CREATE INDEX index_rent  
ON Hostel_Rooms(RentPerMonth);
```

```
SELECT RoomNumber,RentPerMonth  
FROM Hostel_Rooms  
WHERE RentPerMonth>4500;
```

```
DROP INDEX index_rent ON Hostel_Rooms;
```

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x

Limit to 1000 rows

```
6 ON Hostel_Rooms(RentPerMonth);
7
8 • SELECT RoomNumber,
9       RentPerMonth
10 FROM Hostel_Rooms
11 WHERE RentPerMonth > 4500;
12
13 • DROP INDEX index_rent
14 ON Hostel_Rooms;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

RoomNumber	RentPerMonth
B201	5500.00
B203	5500.00

Result Grid
Form Editor
Field Types
Read Only

Hostel_Rooms1 x

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x

Limit to 1000 rows

```
8 • CREATE INDEX idx_student_name
9 ON Hostel_Student(StudentName);
10
11 • SELECT StudentName
12 FROM Hostel_Student
13 WHERE StudentID = 201;
14
15 • DROP INDEX idx_student_name
16 ON Hostel_Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

StudentName
Lokesh

Result Grid
Form Editor
Field Types
Read Only

Hostel_Student 4 Hostel_Student 5 x

2. Create an index on StudentID and display the student whose StudentID is 201.

```
CREATE INDEX index_student
ON Hostel_Student(StudentID);
```

```
SELECT StudentName
FROM Hostel_Student
WHERE StudentID=201;
```

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x

Limit to 1000 rows

```

5  VALUES
6  (151, 'C101', 6000.00);
7
8  SELECT * FROM v_room;
9
10 • DELETE FROM v_room
11   WHERE RoomID = 151;
12
13 • SELECT * FROM v_room;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

RoomID	RoomNumber	RentPerMonth
101	A101	3500.00
102	A102	4500.00
103	B201	5500.00
104	A104	3000.00
105	A105	4500.00
106	B202	4000.00
107	B203	5500.00

v_room 14 v_room 15 x Read Only

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x

Limit to 1000 rows

```

18  VALUES
19  (151, 'C101', 6000.00);
20
21 • SELECT * FROM v_room;
22
23 • DELETE FROM v_room
24   WHERE RoomID = 151;
25
26 • SELECT * FROM v_room;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

RoomID	RoomNumber	RentPerMonth
101	A101	3500.00
102	A102	4500.00
103	B201	5500.00
104	A104	3000.00
105	A105	4500.00
106	B202	4000.00
107	B203	5500.00

v_room 18 v_room 19 v_room 20 v_room 21 x Read Only

RESULT

Thus, VIEW and INDEX were successfully created and used on the Hostel Management System tables.